# A Spiking-Reservoir Reference for Channel-Missingness Evaluation in Affective EEG

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***Abstract*—Affective EEG event-related potentials (ERPs) are noisy, subject-dependent measurements of neural response dynamics. In this setting, robustness rankings can depend as much on the evaluation protocol as on the representation being evaluated. We compare four affective-ERP representations under subject-grouped validation: spectral band-power, ERP-window amplitudes, a fixed leaky integrate-and-fire reservoir summarized by a six-bin spike-count ($\mathrm{BSC}_6$) embedding, and a trained EEGNet, with uncertainty estimated by resampling subjects rather than folds. We isolate two ways a common channel-dropout protocol can distort the measured ranking. First, raw retention rewards representations that sit near the chance floor: the reservoir shows near-total retention at $30\%$ dropout mainly because its clean accuracy is close to the $1/3$ floor, and above-chance retention, while a better summary, is itself unstable near chance. Second, zeroing a dropped channel before standardization is not neutral for a non-centered feature. We show that zero-fill robustness is not invariant to an information-preserving re-centering, so it is partly a property of the coordinate origin rather than of the representation. Across $10$ to $50\%$ dropout and mean, $k$NN, and spatial fills, in-distribution imputation raises the non-centered encoders by up to about $0.10$ balanced accuracy. Under paired subject-level inference, ERP-window features are the strongest fixed encoder under every in-distribution fill, while under zero-fill they are within noise of the reservoir, whose centered embedding barely moves with the fill rule and serves as a diagnostic reference. For EEGNet, channel-dropout augmentation recovers most of the dropout loss (balanced accuracy $0.554$ to $0.674$) while preserving clean accuracy, so its dropout behavior is largely a training choice. The same effect appears on DEAP, where the measured robustness of the same band-power features moves by up to $0.11$ balanced accuracy under the fill rule alone. We recommend reporting absolute accuracy with subject-level uncertainty, stating and varying the fill rule, and reporting trained-model results with and without augmentation.**

***Index Terms*—EEG, event-related potentials, spiking neural networks, reservoir computing, affective computing, neuromorphic computing, ARSPI-Net.**

## I. Introduction

Electroencephalographic event-related potentials (ERPs) provide millisecond-scale measurements of neural response dynamics, but the recorded waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural process [12], [13]. Inter-subject variability, trial averaging, volume conduction, and measurement noise can make clean-split accuracy difficult to interpret. A representation may separate affective conditions under one protocol while preserving little usable structure after amplitude noise, temporal misalignment, or channel loss.

Endpoint accuracy alone also says little about what a representation preserves. This issue is especially relevant for spiking encodings. Fixed recurrent reservoirs and spiking networks provide nonlinear temporal expansions [1], [4], but EEG/ERP studies often treat them primarily as classifiers [9], [15]. In that setting, the reservoir's discrete spike-count features are not usually examined as perturbation-sensitive temporal measurements tied to post-stimulus windows.

The perturbation behavior of affective-ERP reservoir encodings remains underreported, even though preprocessing, acquisition, and electrode-loss choices can materially change EEG representation performance [16]–[18].

This paper evaluates whether channel-dropout robustness rankings for affective-ERP representations are stable under reasonable changes to the scoring protocol. We compare spectral band-power, ERP-window amplitudes, a fixed leaky integrate-and-fire (LIF) reservoir summarized by six post-onset spike-count bins ($\mathrm{BSC}_6$) and compressed for a linear readout, and a trained EEGNet (Fig. 1). The reservoir is not treated as a proposed winner; it is used as a fixed reference whose embedding is nearly zero-centered and whose dynamics are not retrained under perturbation. The contributions are:

1) A subject-grouped comparison of affective-ERP representations under amplitude noise and channel dropout, with uncertainty and encoder contrasts estimated by a subject-level bootstrap rather than by treating folds as independent (Sec. V-A, Tables I–II).
2) Quantification of two protocol artifacts in channel-dropout evaluation: raw-retention inflation near the chance floor, and zero-imputation bias for non-centered features. The imputation effect is also tested on DEAP (Secs. III-E, V-B, V-D).
3) Paired subject-level evidence that in-distribution imputation makes ERP-window features significantly the strongest fixed encoder, whereas under zero-fill they are within noise of the reservoir, and that channel-dropout augmentation, not architecture, accounts for the trained EEGNet's recovered dropout accuracy (Fig. 4, Table II).
4) A fixed-reservoir reference analysis, including a single-bin ablation, to separate the reservoir state-space expansion from temporal bin granularity (Eq. (4)).

Clinical labels and affective categories define the task structure; the representations and their perturbation behavior are the object of study.

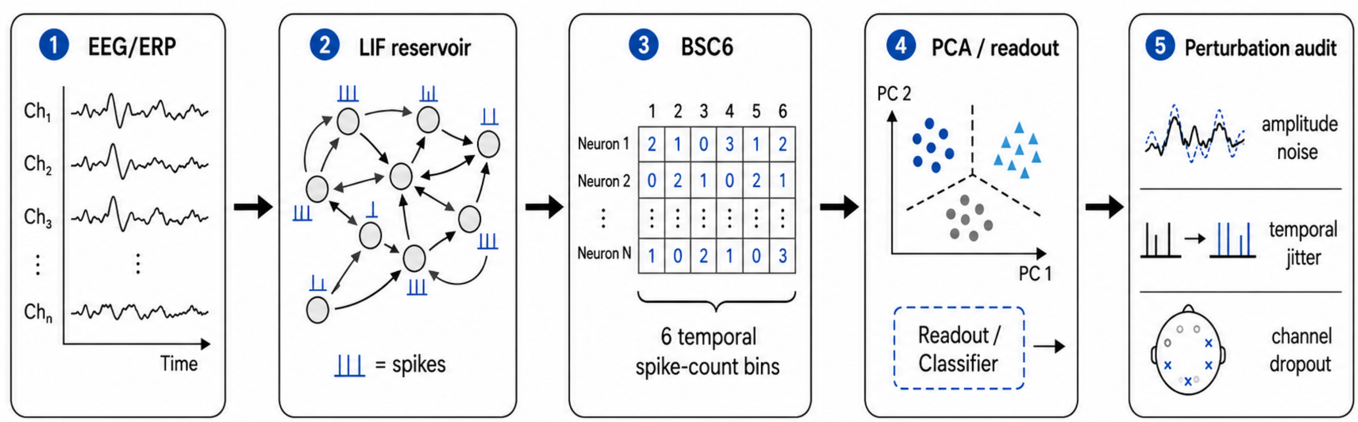


Fig. 1. Reservoir encoding and perturbation-evaluation pipeline. Trial-averaged affective EEG/ERP observations are mapped through a fixed LIF reservoir, summarized by six temporal spike-count bins ($\mathrm{BSC}_6$), compressed for linear readout, and evaluated under amplitude noise, temporal jitter, and channel-dropout perturbations.

## II. Related Work

ERP/EEG analysis commonly uses time-window amplitudes and latencies, Hjorth [10] and wavelet/spectral descriptors [11], and linear models [13]. These features remain attractive because they are interpretable in relation to known ERP components, including the P300 and late positive potential [12], but they can miss nonlinear temporal interactions. Reservoir computing offers a complementary approach: a high-dimensional recurrent dynamical system maps the input to a state trajectory that can be read out by a simple classifier [1]–[4]. Spiking liquid-state machines add event-based timing and count structure [5], [6]. When the recurrent dynamics are fixed, the reservoir can be analyzed as a measurement operator rather than only as a classifier.

Spiking and deep networks are now widely applied to EEG [9], [14], [15]. Much of this literature still emphasizes endpoint accuracy, while preprocessing and acquisition choices receive less systematic treatment [18], [19]. Neuromorphic reservoir features for affective EEG were introduced in prior ARSPI-Net work [7], [8]; here we establish their perturbation and invariance properties and identify two scoring artifacts that distort robustness comparisons.

Robustness to noise, artifacts, and electrode loss is increasingly studied. Dynamic spatial filtering addresses missing or corrupted channels with a learned channel-attention front end [21], and channel-dropout or amplitude-noise augmentation are standard ways to harden trained models [22]. A recent EEG foundation-model benchmark reports that no single model is most robust to all perturbations and that apparent channel-dropout fragility can depend on whether dropped channels are removed or zero-padded [20]. We build on that methodological point by separating two scoring effects: raw-retention inflation for low-accuracy representations and zero-imputation bias for non-centered fixed features. The fixed reservoir is used as a reference case for this analysis because it is training-free and nearly invariant to the imputation rule.

## III. Methods

### A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C\times T}$ denote the trial-averaged ERP observation for subject $s$ and affective condition $c$, with $C$ electrodes (channels) and $T$ temporal samples; observations are stored as $T\times C$ arrays and transposed before encoding. We treat $X_s^{(c)}$ as a noisy observation of an unobserved neural state trajectory:

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \qquad x_t = H\xi_t + \epsilon_t, \tag{1}$$

where $\xi_t$ is the latent neural state, $H$ is a measurement operator induced by scalp observation, and $\epsilon_t$ captures measurement and preprocessing noise. The perturbations below are applied to this observation model (Sec. V-B).

### B. Spiking Reservoir Encoding

For each electrode trace, the signal is passed through a fixed LIF reservoir of 256 neurons with recurrent weights $W$ scaled to spectral radius $\rho = 0.9$, a conventional subcritical reservoir setting, input weights $W_{\mathrm{in}}$, leak parameter $\beta = 0.05$, and firing threshold $\vartheta$. A compact discrete-time representation is

$$v_{t+1} = \beta v_t + W z_t + W_{\mathrm{in}} x_t - \vartheta r_t, \tag{2}$$

$$z_{t+1} = \mathbf{1}[v_{t+1} \geq \vartheta], \tag{3}$$

where $v_t$ is the membrane state, $z_t$ is the binary spike vector, and $r_t$ denotes reset after threshold crossing. The reservoir is fixed during feature extraction; only the downstream readout is trained.

Post-onset activity is summarized by six equal temporal bins over the window $t \in [10, 70]$ in sample units, which at 256 Hz spans approximately 39–273 ms post-stimulus. For reservoir unit $j$ and bin $k$, the binned spike-count code is $b_{j,k} = \sum_{t\in\mathcal{B}_k} z_j(t)$, $k = 1, \ldots, 6$. Per-electrode spike-bin features are compressed to 64 principal components and concatenated across the 34 channels into a trial-level reservoir representation $E_s^{(c)} \in \mathbb{R}^{2176}$. Writing $s_t = z_t \in \{0,1\}^N$ for the reservoir spike state, the encoding is a fixed composition of a nonlinear temporal map, spike-count binning, and the unsupervised projection:

$$E = \phi_{\mathrm{PCA}}\big([b_1, \dots, b_6]\big) = \Phi(X), \tag{4}$$

so $E = \Phi(X)$ is a fixed, label-free representation. The PCA basis is fitted once, unsupervised, over spike-bin vectors pooled across all subjects, hence transductive with respect to subject inclusion (bounded by the fold-local control in Sec. V-A). A single-bin control ($\mathrm{BSC}_1$, total post-onset spike count) uses the identical pipeline. The $t \in [10, 70]$ window is early and compact, excluding the later P300/LPP ranges the ERP-window baseline sees, so the clean comparison is conservative for the reservoir (Fig. 2).

### C. Baselines and Readout

Two classical baselines are evaluated under the same subject-grouped protocol. The band-power baseline (A0) uses spectral band-power features (5 bands $\times$ 34 channels = 170 dimensions). The ERP-window baseline ($\mathrm{A0}_e$) uses mean amplitude in three post-stimulus windows, early (80–200 ms), P300 (250–450 ms), and late positive potential (450–800 ms), per channel ($3 \times 34 = 102$ dimensions). To separate representational robustness from raw dimensionality, we add a dimensionality-matched control: A0 randomly projected to 2176 dimensions (Gaussian random projection, averaged over five random seeds). The band-power, ERP-window, and reservoir representations use a balanced L2-regularized logistic readout to test separability without a high-capacity classifier.

As a trained deep baseline we evaluate EEGNet-8,2 [14] on the raw $34 \times 256$ ERP signal under the identical subject-grouped folds, trained per fold (Adam, 80 epochs) with train-only per-channel standardization. EEGNet ingests the raw signal and has no intermediate feature vector, so its perturbations are applied at the input (comparable to the reservoir's raw-signal recomputation, Sec. V-B), whereas the band-power, ERP-window, and reservoir perturbations are feature-level; the cross-family comparison is therefore approximate.

### D. Perturbation Diagnostics

We separate two experiments. In the feature-coordinate experiment (Sec. V-B), a dropped channel's feature block is suppressed after extraction and refilled by one of four rules, which isolates the effect of the fill convention and the coordinate origin. In the signal-level experiment (Sec. V-C), electrodes are removed from the raw signal and every representation is recomputed from the corrupted signal, so a reservoir channel is zeroed before its PCA mixing rather than after. Formally, a perturbation $\mathcal{P}_\eta$ acts either on the representation, $E \mapsto \mathcal{P}_\eta(E)$, or on the observation model of Eq. (1): amplitude noise augments the measurement noise $\epsilon_t$, channel dropout zeroes rows of the measurement operator $H$, and temporal jitter shifts the sampling index of $x_t$. We apply additive amplitude noise (signal-to-noise ratios), temporal jitter of the alignment window, and channel dropout at increasing electrode-removal rates. Performance is balanced accuracy (BA) and macro-F1.

### E. Robustness Metrics and Two Confounds

We report two robustness summaries. *Raw retention* is the ratio of perturbed to clean BA. Because BA is bounded below by chance ($c = 1/3$ for three balanced classes), raw retention can overstate the stability of representations whose clean performance is already close to chance. We therefore also report *above-chance retention*, $(\mathrm{BA}_{\mathrm{pert}} - c)/(\mathrm{BA}_{\mathrm{clean}} - c)$.

Channel dropout also requires a fill rule for the dropped channel's features. The common zero-fill convention is not neutral after train-only standardization: a zero-valued feature becomes $-\mu/\sigma$, which can be out of distribution for non-centered features. Train-mean imputation maps the dropped feature to the standardized origin. We compare zero-fill with train-mean imputation, then add two in-distribution checks: train-fitted $k$-nearest-neighbor imputation ($k$=10) and a within-sample spatial fill using the mean of retained channels. For EEGNet, perturbations are applied to the raw input, and an augmented training condition is evaluated separately.

*Non-invariance of zero-fill robustness.* Write the standardized readout input as $z = D_\sigma^{-1}(x - \mu)$ with $D_\sigma = \mathrm{diag}(\sigma)$, and let $S$ index the dropped coordinates. Zero-fill sets $x_S = 0$, so the dropped block enters as $z_S = -D_{\sigma,S}^{-1}\mu_S$; train-mean fill sets $x_S = \mu_S$, so $z_S = 0$. The two fills therefore differ by a fixed shift,

$$z_S^{\mathrm{zero}} - z_S^{\mathrm{mean}} = -D_{\sigma,S}^{-1}\mu_S. \tag{5}$$

**Proposition 1.** *Let the clean readout depend on $x$ only through $z$, so it is invariant to a constant offset $x \mapsto x - b$ (train-only standardization guarantees this). Then the clean prediction is unchanged by the re-centering, while the zero-fill dropout input shifts by $\Delta z_S = D_{\sigma,S}^{-1} b_S$. Hence the measured zero-fill dropout accuracy is not a function of the information in $x$ alone; it depends on the coordinate origin, and it is invariant to the fill rule only when $\mu_S = 0$.*

*Proof.* Standardization subtracts the training mean, so the constant $b$ cancels from the clean input and clean accuracy is unchanged. Under the re-centering, a dropped coordinate zero-filled to $x_i = 0$ maps to $z_i = -(\mu_i - b_i)/\sigma_i$, which differs from the original $-\mu_i/\sigma_i$ by $b_i/\sigma_i$. This shift is nonzero for some admissible $b$ unless $\sigma_i^{-1} = 0$, and zero- and mean-fill coincide exactly when $\mu_i = 0$ for all $i \in S$. $\square$

For a linear readout with weights $w$, Eq. (5) changes the dropped-block contribution to the logit by $\sum_{i \in S} w_i \mu_i / \sigma_i$ between the two fills, so the measured gap scales with the non-centeredness $\|D_{\sigma,S}^{-1}\mu_S\|$ of the dropped features. A mean-centered embedding ($\mu = 0$), such as the pooled-PCA reservoir code, is the unique fixed point of the fill rule and is why it serves as a coordinate-independent reference. The bound predicts the subject-level gaps in Sec. V-B: $+0.042$ for ERP-window, $+0.025$ for band-power, and $\approx 0$ for the centered reservoir.

## IV. Experimental Protocol

### A. Dataset and Task

The experiments use trial-averaged affective ERP data from the Stress, Health, and the Psychophysiology of Emotion

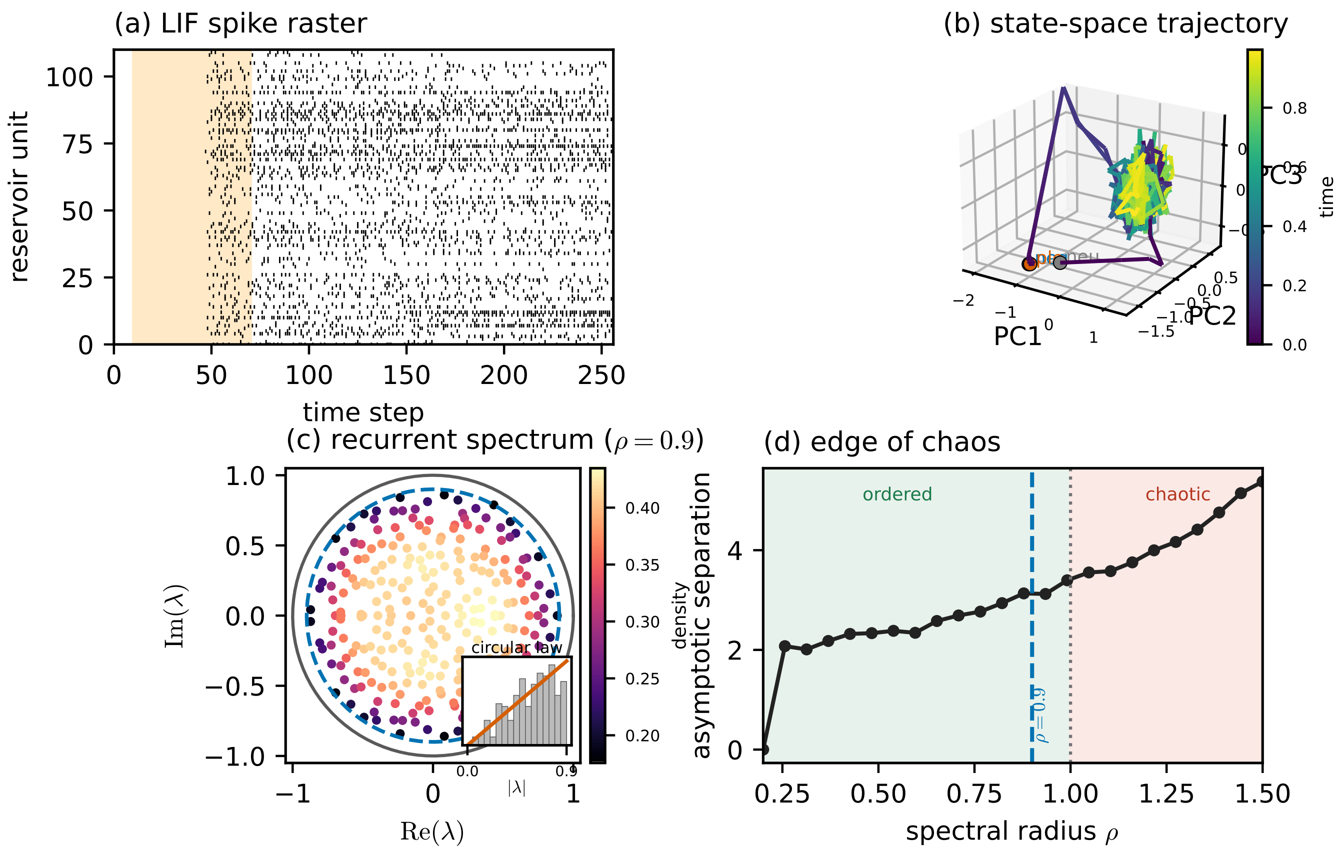


Fig. 2. The fixed LIF reservoir as a dynamical system. (a) Spike raster for one condition (shaded = the $t \in [10, 70]$ analysis window). (b) Population state-space trajectory (first three principal components of the 256-unit membrane state), colored by time, for the three affective conditions; each trajectory settles onto a shared low-dimensional manifold. (c) Eigenvalue spectrum of the recurrent weight matrix, filling the disk of spectral radius $\rho = 0.9$ (dashed) inside the unit circle. (d) Asymptotic state separation as a function of spectral radius; the chosen $\rho = 0.9$ (dashed) sits below the $\rho = 1$ line (dotted).

(SHAPE) project [24], collected by the Laboratory for Clinical Affective Neuroscience at Stony Brook University, with 211 subjects after exclusion of one anomalous subject record. Each subject contributes three affective conditions (negative, neutral, pleasant), yielding 633 subject-condition observations. The ERP traces contain 34 scalp channels downsampled to 256 Hz, with 256 post-stimulus samples per epoch, baseline-corrected relative to onset; the grand averages show the expected late-positive-potential enhancement for emotional over neutral stimuli (Fig. 3). Subject-level data are maintained in a restricted research environment; only aggregate, de-identified results are reported.

### B. Validation

All results use a single protocol: StratifiedGroupKFold (5 folds, seeds 42–46), train-only standardization, and a balanced L2 logistic readout; observations from the same subject never span train and test. Uncertainty is estimated at the level of the subject, not the fold. For each representation we average the out-of-fold class probabilities of every observation across the five repeated partitions, then form 95% confidence intervals (CIs) by resampling the 211 subjects with replacement, keeping all three condition observations of a resampled subject together. Comparisons between encoders or fill rules are reported as paired differences under the same subject resampling. We avoid tests over the 25 folds, which reuse the same subjects and are not independent. Chance is $1/3$ for the three balanced classes, and a label-permutation null lies near that floor (BA $\approx 0.34$). Because the per-observation predictions are averaged over the five partitions, the point estimates below run a few points above the corresponding per-fold means.

## V. Results

### A. Clean Classification and Controls

Table I reports clean three-class balanced accuracy for the three fixed encoders under the subject-level bootstrap. ERP-window features are the strongest fixed encoder (BA 0.629), above band-power (0.490) and the reservoir embedding (0.485); the subject-level intervals for band-power and the reservoir overlap, so they are not clearly separated. A trained EEGNet, evaluated separately on the raw signal, is more accurate still (BA 0.711, per-fold estimate). The fixed encoders therefore carry genuine but modest condition information, and the reservoir is not the most accurate encoder.

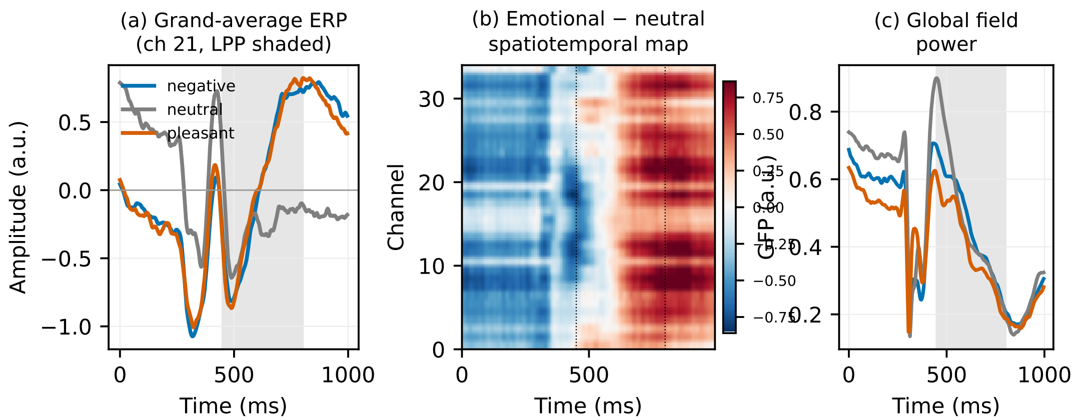


Fig. 3. Trial-averaged affective ERP data (SHAPE cohort; grand averages over 211 subjects). (a) Grand-average waveforms per condition at the most affect-discriminative channel; shaded = the late-positive-potential window (450–800 ms). (b) Spatiotemporal map of the emotional-minus-neutral difference across all 34 channels, showing the sustained LPP enhancement. (c) Global field power per condition.

TABLE I
CLEAN THREE-CLASS BALANCED ACCURACY FOR THE FIXED ENCODERS, WITH SUBJECT-LEVEL BOOTSTRAP 95% CIS (211 SUBJECTS RESAMPLED WITH REPLACEMENT, THREE CONDITIONS EACH). CHANCE IS $1/3$. EEGNET IS A TRAINED RAW-SIGNAL BASELINE, REPORTED SEPARATELY.

| Cfg | Features | BA [95% CI] |
|---|---|---|
| $A0_e$ | ERP-window (102) | 0.629 [0.597, 0.660] |
| A0 | Band-power (170) | 0.490 [0.453, 0.526] |
| A1 | Reservoir $BSC_6$ (2176) | 0.485 [0.447, 0.521] |

*Temporal-binning ablation.* A single-bin total-count control ($BSC_1$) matches $BSC_6$ within the subject-level interval, on clean classification and under perturbation. The reservoir's behavior comes from the fixed state-space expansion, not from six-bin temporal resolution, and we do not claim a temporal-coding benefit for $BSC_6$.

*Transduction and component controls.* A fold-local PCA refit on the training subjects leaves reservoir performance unchanged, and varying the component count over $\{16, 32, 64, 128\}$ does not materially move it. Neither the transductive basis nor the 64-component choice accounts for the results.

### B. Feature-Coordinate Experiment: Two Scoring Confounds

Under amplitude noise the clean accuracy ranking is preserved: the reservoir and EEGNet degrade little, while the windowed classical features fall more. Channel dropout is where the reported conclusion depends on the scoring convention (Table II, Fig. 4).

*Accuracy-floor effect.* Raw retention, the ratio of dropped to clean BA, makes the reservoir look the most robust encoder: at 30% dropout its balanced accuracy barely moves, giving near-total retention. This reflects the ratio, not stability. The reservoir sits close to the $1/3$ chance floor, so it has little above-chance accuracy to lose. Above-chance retention, $(\mathrm{BA}_{\mathrm{pert}} - c)/(\mathrm{BA}_{\mathrm{clean}} - c)$, corrects the obvious floor problem, but its denominator is only about $0.15$ for the reservoir, so it becomes unstable near chance and can exceed one. We therefore treat absolute BA and paired subject-level differences as primary, with retention ratios as secondary diagnostics.

*Zero-imputation effect.* Zeroing a dropped channel's features and then standardizing maps them to $-\mu/\sigma$, an out-of-distribution value when the feature is not centered at zero. Train-mean imputation maps the dropped feature to the standardized origin instead, and Eq. (5) predicts that the gap between the two fills tracks the feature mean. The subject-level estimates match this. At 30% dropout the mean-minus-zero gap is $+0.042$ BA for ERP-window and $+0.025$ for band-power, but only $-0.009$ for the reservoir, whose embedding is centered. In-distribution fills ($k$NN, spatial) raise the non-centered encoders further, by up to about $0.10$ BA, while the reservoir stays within noise of its zero-fill score across all four fills (Fig. 4).

*Encoder comparison.* The reservoir is never the strongest fixed encoder once absolute accuracy is compared. Table II gives the paired subject-level difference between ERP-window and the reservoir at 30% dropout. Under zero-fill the two are within noise (ERP-window minus reservoir = $+0.04$ BA, 95% CI $[-0.00, +0.09]$). Under every in-distribution fill ERP-window is clearly ahead (mean $+0.09$, $k$NN $+0.10$, spatial $+0.10$; all CIs exclude zero). The fill rule does not flip a reservoir lead into an ERP-window lead. Instead it turns a difference that is within noise under zero-fill into a significant one. The reservoir's role is as a centered reference whose score is insensitive to the fill rule, which is what makes the effect in the other encoders visible.

*Training effect.* EEGNet ingests the raw signal, so channel dropout is applied at its input. Unaugmented, it drops to $0.554$ BA at 30% dropout. Retraining with channel-dropout and amplitude-noise augmentation recovers it to $0.674$ while preserving clean accuracy ($0.703$). For a trained model, dropout behavior is set mainly by the training procedure, not by the architecture (Fig. 5b).

TABLE II

CHANNEL DROPOUT AT 30% (FEATURE-COORDINATE EXPERIMENT): SUBJECT-LEVEL BALANCED ACCURACY FOR THE FIXED ENCODERS UNDER FOUR FILL RULES, AND THE PAIRED DIFFERENCE ERP-WINDOW MINUS RESERVOIR UNDER THE SAME SUBJECT RESAMPLING. A POSITIVE DIFFERENCE WHOSE 95% CI EXCLUDES ZERO MEANS ERP-WINDOW IS SIGNIFICANTLY AHEAD. PER-ENCODER 95% CIS ARE ABOUT $\pm 0.035$ (TABLE I SCALE).

| Fill | Band-pow. | ERP-win. | Reserv. | ERP−Res [95% CI] |
|---|---|---|---|---|
| Zero | 0.455 | 0.536 | 0.494 | +0.04 [−0.00, +0.09] |
| Mean | 0.480 | 0.578 | 0.485 | +0.09 [+0.05, +0.14] |
| $k$NN | 0.507 | 0.610 | 0.513 | +0.10 [+0.06, +0.14] |
| Spatial | 0.479 | 0.589 | 0.493 | +0.10 [+0.06, +0.14] |

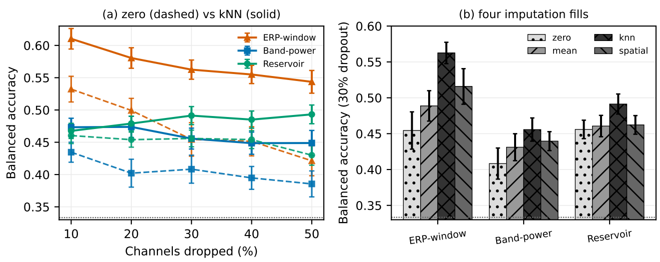


Fig. 4. Channel-dropout performance under different imputation rules. (a) Dropout-rate curves under zero (dashed) and kNN (solid) imputation. The reservoir's centered embedding changes little across fills, whereas ERP-window and band-power features improve under in-distribution fills. (b) At 30% dropout, zero, mean, kNN, and spatial fills move the non-centered encoders by 0.05–0.11 BA, while the reservoir changes only modestly. Dashed line = chance.

### C. Signal-Level Channel Removal

The feature-coordinate experiment suppresses a channel after extraction. We also ask the operational question: what happens when an electrode is actually missing before extraction? Here electrodes are removed from the raw signal and every representation is recomputed from the corrupted signal, with the reservoir PCA refit on the clean training fold. A removed electrode contributes zero band-power, zero ERP amplitude, and, because a zero drive from a zero reservoir state produces no spikes, a zero spike-count block that then enters the PCA mixing. Table III reports subject-level balanced accuracy at each removal level.

Two points stand out. ERP-window features remain the strongest fixed encoder under real channel loss, at every level. And the reservoir's insensitivity to the fill rule does not carry over: once a channel is zeroed before the PCA mixing, the reservoir degrades with removal like the others, reaching 0.395 BA at 50%, near chance. The imputation-invariance seen in the feature-coordinate experiment is therefore a property of the centered feature space, not physical robustness to missing electrodes. EEGNet, which also ingests the raw signal, is the most accurate under channel loss once augmentation is included (Sec. V-B).

TABLE III

SIGNAL-LEVEL CHANNEL REMOVAL: SUBJECT-LEVEL BALANCED ACCURACY [95% CI] WITH EVERY REPRESENTATION RECOMPUTED FROM THE CORRUPTED SIGNAL. CHANCE IS $1/3$.

| Removed | Band-power | ERP-window | Reservoir |
|---|---|---|---|
| 0% | 0.494 | 0.629 | 0.480 |
| 10% | 0.476 | 0.602 | 0.471 |
| 30% | 0.417 | 0.536 | 0.474 |
| 50% | 0.450 | 0.474 | 0.395 |

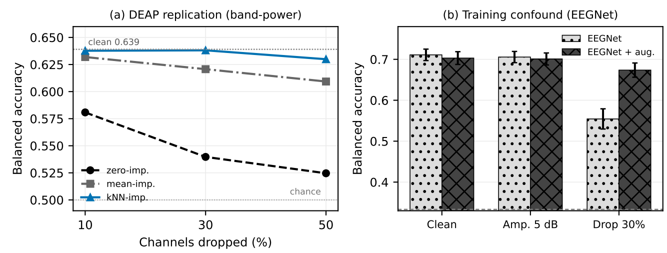


Fig. 5. (a) External check on DEAP [23] (32 subjects, within-subject binary valence, band-power; clean BA 0.639). Channel-dropout BA changes substantially with the imputation rule. (b) EEGNet channel-dropout performance improves from 0.554 to 0.674 when channel-dropout augmentation is used during training, with clean accuracy preserved.

### D. External Replication on an Independent Affective Dataset

We next test whether the effect is specific to SHAPE. On DEAP [23] (32 subjects, band-power features, binary valence, within-subject 5-fold), the same pattern appears and is larger (Fig. 5a): at 30% dropout the same features score BA 0.540 under zero-fill but 0.621–0.638 under mean or $k$NN. With clean BA 0.639, the fill rule alone moves measured accuracy under dropout by up to 0.11 BA, on a second dataset, task, and montage. We report this as external validation of the fill-rule effect, not as a replication of the full comparison.

## VI. DISCUSSION

The main result is methodological. Channel-dropout robustness estimates change under two common reporting choices: raw retention can favor models close to the chance floor, and zero-imputation can penalize non-centered features after standardization. For trained models, augmentation is a third factor. After accounting for these choices, the fixed reservoir should not be described as the most channel-robust encoder. Under paired subject-level inference, ERP-window features are significantly the strongest fixed encoder under every in-distribution fill and are within noise of the reservoir only under zero-fill, and the augmented EEGNet is the most accurate baseline under channel loss. The reservoir remains useful as a fixed reference because its centered embedding changes little across fill rules, which makes the effect in the other encoders visible. That invariance is specific to the feature space: when electrodes are removed at the signal level, the reservoir degrades like the other encoders, so it should not be read as robustness to missing channels.

These results suggest a minimal reporting standard: report absolute accuracy with subject-level uncertainty, state and vary the channel-fill rule, and report trained-model results with and without augmentation. None of these require a new model, yet each can change the comparison.

**Limitations.** The primary cohort is a single trial-averaged affective-ERP dataset. The imputation effect follows from the standardization algebra in Eq. (5) and is observed across three fill rules, five dropout rates, and the independent DEAP check, but paradigm-matched multi-cohort ERP validation would still strengthen the claim. The signal-level experiment recomputes every fixed encoder under matched channel removal; EEGNet is reported from per-fold estimates rather than the subject-level bootstrap, so its comparison to the fixed encoders is approximate. Adversarial attacks and montage shifts are not evaluated here. Graph topology and structure–function coupling are outside the scope of this paper.

## VII. Conclusion

Channel-dropout robustness rankings for affective-ERP representations depend on the scoring protocol. Raw retention can overstate robustness near the chance floor, zero-imputation can distort the behavior of non-centered features after standardization, and trained models need to be evaluated with their augmentation state made explicit. In this study, in-distribution imputation makes ERP-window features significantly the strongest fixed encoder under subject-level inference, and the same effect appears on DEAP. Future work should test the same reporting protocol on paradigm-matched, multi-cohort ERP data.

**Ethics.** The SHAPE study was collected by the Stony Brook University Laboratory for Clinical Affective Neuroscience under that laboratory's Institutional Review Board oversight. DEAP is a publicly released dataset governed by its original collection protocol [23].

**Data Availability.** For information on the SHAPE study dataset, contact the Laboratory for Clinical Affective Neuroscience at Stony Brook University [24]. DEAP is public [23].

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